# Dominus OS

## A Human-Governed Operating System for AI Workforces

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### Abstract

AI tools are accelerating, but most organizations still lack durable governance for automated work: authority is unclear, audit trails are incomplete, context is fragmented, and workflows fail quietly. These gaps turn capability into operational risk.

Dominus OS is a human-governed operating system for AI workforces. It provides a consistent structure for orchestrating agents and automated workflows under explicit authority controls, structured memory, and verifiable observability. The objective is not more automation, but trustworthy automation: systems that remain accountable to human intent, resilient under stress, and auditable when outcomes matter.

This paper describes the problem, presents the governing thesis, outlines the conceptual architecture, and documents both the current state of the system and the public milestones ahead.

**Keywords:** AI governance; agent orchestration; human-in-the-loop; auditability; observability; workflow integrity; memory architecture; operational safety; multi-tenant systems; authority control

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### 1. The Problem: AI Without Governance Becomes Liability

The current generation of AI tools has made it remarkably easy to automate individual tasks. A language model can draft an email, summarize a document, or generate a report in seconds. But the ease of automating a single task masks a harder problem: what happens when dozens or hundreds of automated tasks run simultaneously, across business functions, touching live data, making decisions that compound over time?

Without governance, automated systems accumulate risk in ways that are difficult to detect and expensive to unwind. The following failure modes are not theoretical. They are observable today in organizations deploying AI at any meaningful scale.

**Authority ambiguity.** When an AI agent sends an email, updates a customer record, or modifies a document, who authorized that action? In most current architectures, the answer is unclear. The agent was given access to a tool, and it used it. There is no record of which human approved the action, under what scope, or with what constraints. In a regulated environment, this ambiguity alone can be disqualifying. In any environment, it makes root-cause analysis after an incident substantially harder. Consider a sales assistant that sends a pricing proposal to a prospect. If the price was wrong, who is accountable -- the person who configured the agent, the person who triggered the workflow, or the agent itself? Without explicit authority records, the answer is a committee meeting, not a log entry.

**Silent failure.** Most AI workflows fail without raising an alarm. An API call returns an error, and the agent moves on. A data source goes stale, and the agent continues to use outdated information. A scheduled job stops running, and no one notices until a downstream process breaks days later. Silent failure is especially dangerous because it erodes trust invisibly. The system appears to be working. Reports are generated, tasks are marked complete, dashboards show green. But the underlying data is wrong, the actions were incomplete, or the process skipped a critical step. By the time the failure surfaces, the damage has compounded. An operations team relying on an AI-generated daily briefing may not realize the briefing stopped pulling from a key data source three weeks ago. Every decision made in those three weeks was informed by incomplete information.

**Unverifiable outputs.** When an AI system produces a result -- a recommendation, a summary, a decision -- can you trace how it arrived at that result? In most deployments, the answer is no. The model received some input, produced some output, and the intermediate reasoning is opaque. This is not merely an academic concern about explainability. It is a

practical concern about trust. If a compliance officer asks why a particular customer was flagged, or why a particular document was prioritized, the system should be able to produce the chain of evidence: what data was consulted, what logic was applied, what authorizations were in effect, and what the intermediate steps were. Without this, AI outputs are assertions, not evidence.

**Fragmented context.** As organizations deploy AI across multiple business functions, each function tends to develop its own isolated set of tools, data sources, and workflows. The sales team uses one AI assistant, the operations team uses another, and the executive team uses a third. None of them share context. This fragmentation means that intelligence gathered in one domain cannot inform decisions in another. A customer support interaction that signals churn risk never reaches the account management workflow. A delayed project timeline never surfaces in the client communication queue. The organization has data, but the data is trapped in silos that the AI systems cannot bridge. Each new tool added to the stack increases this fragmentation, creating a governance surface area that grows faster than the organization's ability to manage it.

**No control plane.** Perhaps most critically, there is no unified layer from which a human operator can observe, manage, and intervene in all running AI processes. Each tool has its own dashboard, its own logs, its own failure modes. There is no single view that answers the question: "What is every AI system in my organization doing right now, and can I stop any of it with one action?" The absence of a control plane means that as AI adoption scales, the operator's ability to maintain oversight does not scale with it. More automation produces more activity, more logs, more dashboards, and less coherent understanding.

**The compounding problem.** These failure modes do not exist in isolation. They compound. Authority ambiguity makes silent failures harder to diagnose. Silent failures make outputs harder to verify. Unverifiable outputs make fragmented context more dangerous. And without a control plane, the compounding is invisible until something breaks publicly.

**The gap between demo and production.** There is a well-documented pattern in AI deployment: the demonstration is impressive, and the production deployment is fragile. A system that works well in a controlled setting -- with a single user, clean data, and predictable inputs -- behaves differently when exposed to real-world conditions: concurrent users, stale data, edge cases, network failures, and the accumulated drift of many small decisions over time. Governance is what bridges this gap. It is the infrastructure that makes a system not just capable, but reliable, accountable, and safe to operate at scale.

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## 2. Thesis: Governed Execution as the Missing Layer

The next era of AI will be defined less by raw intelligence and more by governed execution. The bottleneck in most organizations is no longer "Can AI do this task?" but rather "Can AI do this task in a way that is authorized, observable, controlled, contextualized, and reversible?"

This is a meaningful distinction. The prevailing direction in the AI tooling ecosystem is toward greater autonomy: more tools, faster execution, longer chains of reasoning, less human involvement. This trajectory optimizes for capability. Dominus OS optimizes for accountability.

The difference is not philosophical. It is architectural. A system optimized for capability asks: "How can the agent accomplish the goal with the fewest constraints?" A system optimized for accountability asks: "How can the agent accomplish the goal while preserving the operator's ability to understand, verify, and intervene at every step?" These are different design problems, and they produce different architectures.

Most existing agent frameworks treat governance as a feature to be added later -- a logging layer, a permissions check, a human-approval step bolted onto an otherwise unconstrained execution pipeline. Dominus OS treats governance as the foundation. Authority, observability, side-effect control, memory structure, and operator sovereignty are not features. They are invariants -- properties that the system guarantees regardless of what agents are doing or how complex the workflow becomes.

The five governing invariants of Dominus OS are:

**1. Authority is explicit.** Every action that modifies state, sends a communication, or commits a resource is tied to an explicit authorization. This authorization specifies who approved the action, what scope it covers, and what constraints apply. Authority is not inferred from access. Having a credential to an API does not constitute permission to use it. Authority is granted, scoped, and recorded as a first-class element of the system's execution model. This means that at any point, for any action the system has taken, it is possible to answer the question: "Who authorized this, and under what terms?"

**2. Execution is observable.** Every operation the system performs -- whether initiated by a human operator or by an autonomous schedule -- produces a durable record of what happened, what data was involved, what the outcome was, and how long it took. Observability is not optional logging that can be disabled for performance. It is a structural property of the execution layer. The system does not execute work that it cannot observe. This record is not just for debugging. It is for accountability: the ability to reconstruct, after the fact, exactly what happened and why.

**3. Side effects are controlled.** The system distinguishes between operations that read data and operations that change the world. A query that retrieves information is categorically different from an action that sends an email, updates a record, or triggers an external process. This distinction is enforced architecturally, not by convention. Write operations pass through approval gates that are appropriate to their risk level. An agent can plan and reason freely. But the moment it attempts to act on the world, the governance layer intervenes to ensure the action is authorized, scoped, and recorded.

**4. Memory is structured.** The system maintains context across sessions, domains, and time horizons through a structured memory architecture. This is not a conversation history or a vector database. It is a layered system that distinguishes between ephemeral working context (what is relevant to the current task), persistent operational memory (what the organization knows and has verified), and archival records (what happened, preserved immutably). Each layer has its own retention policy, access controls, and provenance tracking.

**5. Operators remain sovereign.** The human operator can, at any time, observe all running processes, halt any process, revoke any authority, and audit any outcome. This sovereignty is not a dashboard feature. It is an architectural guarantee. The system is designed so that no automated process can outrun the operator's ability to intervene. Sovereignty means that AI serves as an amplifier of human judgment, not a replacement for it.

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## 3. Conceptual Architecture

The Dominus OS architecture is organized into four interdependent layers. Each layer has a distinct responsibility, and the boundaries between them are enforced structurally.

**The governance plane** is the authority layer. It defines who can do what, under what conditions, and with what oversight. The governance plane evaluates every proposed action against the authority model before the action is permitted to execute. It maintains the canonical record of permissions, approval chains, and authority boundaries. The governance plane does not execute work. It authorizes work.

**The execution plane** is where agents and automated workflows perform tasks. The execution plane handles tool invocation, API calls, data transformations, and workflow orchestration. Critically, the execution plane cannot act without authorization from the governance plane. Every tool call, every external API invocation, every state mutation passes through the governance boundary. The execution plane is designed to be extensible -- new tools and integrations can be added without modifying the governance model -- but every extension is subject to the same authority controls.

**Structured memory** provides context persistence across sessions, domains, and time horizons. Memory is not a single store. It is a layered architecture with distinct characteristics at each level: working memory for active task context, operational memory for persistent organizational knowledge, and archival memory for immutable historical records. Each memory layer has its own write policies, retention rules, and access controls.

**The observability surface** captures and presents the full execution record. This includes not just success and failure states, but the causal chain that led to each outcome: what was attempted, what was authorized, what data was consulted, what the intermediate results were, and what the final state is. The observability surface is designed for operators, not developers. It presents information at the level of business actions, not system logs.

These four layers interact in a specific pattern. When work is initiated -- whether by a human operator or by an autonomous schedule -- the execution plane proposes actions. Each proposed action is evaluated by the governance plane against the current authority model. Authorized actions are executed, and the results are recorded by the observability surface and persisted in structured memory. Unauthorized actions are blocked, and the block is itself recorded. The operator can observe this entire cycle in real time and intervene at any point.

The architecture is multi-tenant from the ground up. Tenant boundaries are enforced at the data layer, the execution layer, and the governance layer. One tenant's agents, data, memory, and authority model are invisible to another tenant's processes. This is not a feature added after initial development. It is a structural property of the system.

A critical design principle is the separation of reasoning and acting. The system can plan, analyze, and reason without restriction. An agent can evaluate options, draft proposals, and simulate outcomes freely. But the moment the system attempts to act -- to send, publish, modify, or commit -- the governance plane intervenes. Planning is unconstrained. Execution is governed. This separation ensures that the system's analytical capabilities are not diminished by its safety controls, while its real-world impact remains under explicit human authority.

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## 4. Authority and Approvals

Authority in Dominus OS is not a binary switch between "allowed" and "denied." It is a graduated model with three distinct modes, each appropriate to different risk levels and operational contexts.

**Read-only mode** permits an agent to query data, retrieve information, and perform analysis, but prohibits any modification of external state. An agent operating in read-only mode can research a topic, compile a report, or analyze a dataset, but it cannot send an email, update a record, or trigger an external process. Read-only mode is the default for new integrations and the fallback when authority is ambiguous.

**Guarded mode** permits an agent to prepare actions but requires explicit human approval before execution. In guarded mode, the system presents the operator with a complete description of the intended action: what will be done, what data will be affected, what external systems will be touched, and what the expected outcome is. The operator reviews this proposal and either approves, modifies, or rejects it. Only after approval does the action execute. Guarded mode is the standard operating mode for actions with meaningful consequences -- communications sent to external parties, modifications to customer records, financial commitments, or any action that would be difficult to reverse.

**Autonomous mode** permits an agent to execute pre-authorized actions without per-action approval. Autonomous mode is reserved for well-understood, repeatable operations where the risk profile is known and the acceptable boundaries are clearly defined. Even in autonomous mode, every action is recorded, and the operator retains the ability to halt execution and revoke autonomous authority at any time. Autonomous mode is not "unmonitored mode." It is "pre-approved mode with continuous oversight."

The transition between modes is itself governed. Escalating from read-only to guarded, or from guarded to autonomous, requires explicit operator authorization. De-escalation -- moving from autonomous to guarded, or from guarded to read-only -- can be triggered by the operator at any time, and also occurs automatically when the system detects conditions outside the expected operating parameters.

A key architectural decision is that authority boundaries are structural, not behavioral. They are enforced by the system's architecture, not by instructions given to the AI model. An agent cannot be prompted or instructed to bypass an authority gate. The gate exists at the infrastructure level, between the agent's intent and the execution layer. This means that even if an AI model behaves unexpectedly -- hallucinating authority it does not have, or reasoning itself into an action it should not take -- the governance plane prevents the unauthorized action from executing. The model's behavior is bounded by architecture, not by the model's own compliance with instructions.

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## 5. Observability: Trust Requires Evidence

Trust in automated systems is not a feeling. It is a function of evidence. An operator trusts a system to the extent that the system can demonstrate what it did, why it did it, and what the consequences were. Observability in Dominus OS is designed to produce this evidence continuously and durably.

The minimum expectations for operator-grade observability are:

- **Every action produces a record.** Not just successful actions. Failed actions, blocked actions, retried actions, and skipped actions all generate entries in the activity ledger. The absence of a record is itself a signal.
- **Records include full attribution.** Each record identifies the agent that performed the action, the human who authorized it, the data sources consulted, the tools invoked, and the timestamp of each step. Attribution is not approximate. It is precise and traceable.
- **Failure is surfaced, not swallowed.** When an operation fails, the failure is reported to the operator through the observability surface. The system does not retry silently, skip the step, or substitute a default. Failure is treated as information that the operator needs, not as an inconvenience to be hidden.
- **Causal chains are reconstructable.** For any outcome the system has produced, it is possible to trace backward through the full sequence of events that led to that outcome: the initial trigger, the authority under which it executed, the data it consulted, the intermediate decisions, and the final result. This is not a debugging exercise. It is a standard operating capability available to the operator at any time.

"Operator-grade" observability is a specific standard. It means that a non-technical operator -- a business owner, a department head, a compliance officer -- can understand what the system did without reading code or parsing raw logs. The observability surface presents information in terms of business actions ("Drafted follow-up email to client," "Updated lead score based on new signal," "Halted outreach sequence due to bounce"), not system events ("POST /api/v1/messages returned 200," "Row updated in table X"). The operator sees the work, not the plumbing.

Freshness is a distinct dimension of observability. The system tracks not only when data was last fetched, but when it was last verified against its source. A contact's email address may have been fetched yesterday, but if it was last verified three months ago, the system surfaces that distinction. Staleness is treated as a risk factor, not a metadata curiosity. Operators can set freshness thresholds, and the system flags data that has exceeded those thresholds before it is used in a consequential action.

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## 6. Memory: Context Without Confusion

AI systems that lack structured memory are forced to rediscover context every session. They ask the same questions, make the same mistakes, and lose the organizational knowledge that accumulates over time. But AI systems with unstructured memory -- a large, undifferentiated pool of past interactions -- create a different problem: context confusion, where outdated information contaminates current decisions, and the system cannot distinguish between what it knows, what it used to know, and what it was told by an unreliable source.

Dominus OS addresses this through a three-layer memory architecture.

**Working memory** holds the context relevant to the current task or session. It includes the active conversation, the data currently being processed, and the intermediate results of in-progress work. Working memory is ephemeral by design. It is scoped to a specific task and discarded or archived when the task completes. This prevents context bleed between unrelated tasks and ensures that each new task starts with a clean, relevant context rather than the accumulated residue of previous work.

**Operational memory** holds persistent organizational knowledge: verified facts, learned preferences, established procedures, and cross-domain intelligence. Operational memory persists across sessions and is shared (within tenant boundaries) across agents and workflows. When an agent learns that a particular client prefers communication by email rather than phone, that preference is stored in operational memory and available to any future workflow involving that client. Operational memory is curated. Not everything an agent encounters is promoted to operational memory. Information must meet relevance and reliability thresholds before it is persisted.

**Archival memory** holds the immutable historical record: what happened, when, by whom, and with what outcome. Archival memory is append-only. It cannot be modified or deleted. It serves as the system's institutional memory and as the evidence base for audits, compliance reviews, and post-incident analysis. Archival memory is the foundation of the system's accountability guarantees.

Provenance is a first-class property of every memory write. When information is stored -- at any layer -- the system records where the information came from, when it was written, what agent or human created it, and how it is scoped. This means that when the system uses a piece of information to make a decision, the operator can trace that information back to its origin and evaluate its reliability.

The system also maintains a clear distinction between information that was generated by an AI agent and information that was verified by a human. AI-generated information is useful but provisional. Human-verified information carries a higher confidence level and is treated accordingly in downstream decisions. This distinction is not a label applied after the fact. It is a structural property of the memory system, enforced at write time and visible at read time.

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## 7. Use Cases

The following scenarios illustrate how governed execution changes outcomes in practice. Each describes a common operational situation and shows how the Dominus OS approach -- explicit authority, structured memory, and operator-grade observability -- produces a different result than unstructured automation.

**Sales and CRM Operations.** A professional services firm uses AI to support its business development pipeline through a governed CRM. The system researches prospects, enriches contact records with publicly available information, drafts personalized outreach messages, and prepares follow-up sequences based on engagement signals. A composite lead scoring engine combines multiple signal types -- email engagement, meeting frequency, response timing, sentiment analysis -- into a single prioritized view. Critically, the system operates in guarded mode for all external communications: it drafts and prepares, but sending requires explicit authorization from the assigned salesperson. Automated truth-checking runs continuously, flagging stale data, overdue follow-ups, and leads that have gone cold. Every touchpoint -- research conducted, data enriched, message drafted, message sent, response received -- is logged with full attribution, creating a complete record of the client relationship from first contact onward. Email performance analytics track open rates, response rates, and engagement patterns across the entire pipeline, with the data refreshed automatically under governed schedules. When a salesperson is out of office, the system continues to prepare but does not send. When a lead is reassigned, the full context transfers with it, including the reasoning behind previous outreach decisions. The operator can see, at any moment, exactly where every prospect stands and what the system has done on their behalf.

**Executive Control Tower.** An operations leader oversees an organization running AI-assisted processes across multiple business functions. The control tower provides a unified view of all active processes: what is running, what is queued, what has completed, and what has failed. Each process is displayed with its current status, its authority level, its most recent action, and its next scheduled action. The operator can drill into any process to see its full execution history, including every decision point, every data source consulted, and every authorization granted or denied. If an anomaly is detected -- an unusual pattern of failures, an unexpected spike in activity, or a process that has been running longer than expected -- the operator can halt that process with a single action. The halt propagates immediately, and the system records the halt, the reason, and the operator who initiated it. The control tower is not a dashboard of metrics. It is a live operational view with direct intervention capability.

**Compliance-Sensitive Workflows.** A firm operating in a regulated environment uses AI to process incoming documents, route them through approval chains, and maintain the evidence trails required by its regulatory obligations. The system extracts structured data from unstructured documents, classifies them according to the firm's taxonomy, and routes them to the appropriate approvers based on content type, value thresholds, and regulatory requirements. Every step in this chain is recorded immutably: who submitted the document, how it was classified, who approved it, when approval was granted, and what the downstream actions were. Tenant isolation ensures that one client's documents are never visible to another client's processes, even when both clients are served by the same system. When an auditor requests evidence of the approval chain for a specific transaction, the system can produce the complete record -- not a reconstruction from fragmented logs, but a continuous, contemporaneous audit trail that was generated as the work was performed.

**Multi-Domain Operations.** An organization runs AI-assisted processes across sales, operations, finance, and client services. Each domain operates within its own authority boundaries: the sales system cannot modify financial records, and the finance system cannot send client communications. But within those boundaries, cross-domain intelligence flows through structured memory. A signal detected in client services -- a change in communication frequency, a shift in tone, a new requirement mentioned in passing -- is captured in operational memory and surfaced to the relevant domain when it

becomes actionable. The sales team sees the signal as context for their next interaction. The operations team sees it as input for resource planning. The intelligence crosses domain boundaries. The authority to act on it does not. This architecture allows the organization to benefit from integrated intelligence without creating the risk of one domain's automation interfering with another domain's operations.

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## 8. Roadmap and Current State

Transparency about what exists today and what remains to be built is essential to credibility. The following is a candid assessment of the system's current state and near-term direction.

### What Exists Today (Production)

Dominus OS is not a concept paper. The following components are operational and serving real workloads:

- **Execution kernel.** A server-side orchestration kernel managing execution across more than 150 API endpoints, with real-time event distribution via WebSocket, structured job scheduling, and multi-tenant data isolation. The kernel routes objectives to virtual employees, enforces authority gates on every write operation, and maintains the canonical process ledger.
- **Virtual employees.** Seven distinct virtual employees executing real work -- email management, CRM operations, document processing, operational intelligence, compliance enforcement, and cross-domain coordination -- each operating within governed authority envelopes that define what they can read, what they can modify, and what requires human approval. Each virtual employee has a published manifest describing its tools, authority level, and work surface.
- **Native operator applications.** A macOS desktop application built in Swift providing operator-level visibility into all running processes, tab-based task management for concurrent workflows, live WebSocket event streams, and direct halt authority over any active process. A companion iOS application extends operator visibility and intervention capability to mobile, including widget-based status monitoring.
- **Governed CRM.** A purpose-built web CRM application serving field teams with lead management, composite scoring, email attribution tracking, and signal intelligence. Unlike traditional CRMs, every record modification passes through the same governance kernel -- authority gates, audit trails, and attribution chains apply to CRM operations identically to all other system actions. The CRM includes automated lead scoring, follow-up enforcement, stale detection, and compliance truth-checking, all running as governed scheduled jobs.
- **Forge design system.** A unified design language -- typography, color tokens, spacing, and component patterns -- shared across the macOS and iOS operator applications. Forge ensures that the operator experience is consistent, accessible, and structurally tied to the governance layer rather than styled as an afterthought. Every interface element maps to a semantic token, enabling systematic updates across all operator surfaces.
- **Autonomous scheduled operations.** More than 40 scheduled jobs running under governance, each with defined authority levels, failure handling, and audit trails. These include CRM operations (lead scoring, truth checks, follow-up enforcement, stale detection), operational intelligence (briefing generation, performance analysis), and system maintenance tasks.
- **Immutable activity ledger.** A complete, append-only record of all system activity with full attribution chains linking every action to its authorization, its data sources, and its outcome.
- **Cross-domain signal intelligence.** A signal bus architecture that connects intelligence gathered in one business domain to decisions made in another, within tenant and authority boundaries. Signals detected in CRM (engagement patterns, sentiment shifts) surface in operational briefings. Signals from email analysis inform lead scoring. The intelligence crosses domain boundaries; the authority to act on it does not.
- **Integration layer.** Webhook-based integration with third-party services, subject to the same governance controls as internally initiated actions. GitHub repository operations, calendar management, and external API calls all pass through the same authority gates.

### Near-Term Public Milestones

- Publish this thesis and controlled public artifacts documenting the governance model.
- Demonstrate authority controls and the approval workflow end-to-end in a public-facing context.

- Ship operator-grade observability features with surfaced failure reporting and measurable data freshness indicators.
- Expand the library of tool adapters and integrations while preserving governance invariants -- every new integration subject to the same authority, observability, and memory controls as existing ones.
- Continue developing the native iOS application, extending operator visibility, widget-based monitoring, and intervention capability to mobile.

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## Conclusion

The central claim of this paper is that the most important unsolved problem in applied AI is not capability but governance. The ability to make AI systems do more is advancing rapidly. The ability to make AI systems do more while remaining accountable, observable, and under human control is not advancing at the same pace. This gap represents the primary barrier between AI as a promising tool and AI as a trustworthy operational foundation.

Dominus OS addresses this gap by treating governance as architecture, not afterthought. Authority is explicit and structurally enforced. Execution is observable at operator grade. Side effects are controlled through graduated approval models. Memory is structured with provenance and freshness tracking. And the human operator retains sovereign control -- the ability to observe, intervene, halt, and audit at any time, regardless of the system's scale or complexity.

The system described in this paper is not a projection. Its core components are operational and serving real workloads today. The work ahead is to extend its reach, harden its guarantees, and demonstrate its model publicly. The measure of success will not be what the system can do, but what the system can be trusted to do -- reliably, accountably, and under human governance.

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*Dominus Foundry builds AI systems that organizations can govern, audit, and trust. For inquiries, contact mark@dominusfoundry.com.*